

Soc. Stats. II Moderating and Mediating Variables

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Executive Summary

In this assignment, I investigate test the following five hypotheses:

1. The % of a county's population identifying as Hispanic (**s_pcthsp**) predicts crime rate (**indexrt**) in a positive direction.
2. The metropolitan status of a county (**metro00 = 1**) predicts crime rate in a positive direction.
3. The % of a county's population in poverty (**s_pctprv**) predicts crime rate in a positive direction.
4. The % of a county's population identifying as Hispanic predicts metropolitan status in a positive direction.
5. The % of a county's population in poverty predicts metropolitan status in a positive direction.

First, a conceptual model illustrating the potential mediating effect of metropolitan status is presented (*Figure 1*). This model is then justified through a correlation plot (*Figure 2*), wherein it is seen that % in poverty and % Hispanic are, in-fact, correlated with crime rate. However, the strongest correlation between all variables is with metropolitan status. Therefore, while direct effects of the percentage variables on crime rate are observed, untangling the indirect effects of metropolitan status as a mediating variable necessary.

To illuminate the direct, indirect, and total effects of the variables of interest and interrogate the five above hypotheses, I utilize multiple linear regression models. The first model regresses % Hispanic (*a*) and % in poverty (*d*) on metropolitan status. This model is then followed by a second model that regresses % Hispanic (*b*), % in poverty (*e*), and metropolitan status (*c*) on crime rate. The direct effect of the % Hispanic ($b^* = 0.026$) can be found in the labelled coefficient (*b*) and the direct effect of the % in poverty ($b^* = 0.040$) can be found in the labelled coefficient (*e*). Indirect effect of the % Hispanic ($b^* = 0.073$) is found by

multiplying a by c , and the indirect effect of the % in poverty ($b^* = 0.087$) is found by multiplying d by c . Total effect of the % Hispanic ($b^* = 0.099$) is found by computing $b + (a * c)$, and the total effect of the % in poverty ($b^* = 0.127$) is found by computing $e + (d * c)$. These regression results are found in *Table 1*, and provide strong support for conceptualizing metropolitan status as a mediating variable that dismisses potentially prejudiced beliefs on higher crime, and instead supports turning attention toward metropolitan status, and future analysis can further illuminate the influence of socioeconomic status on crime rate within more populous metropolitan areas.

These results are presented in *Figure 3* displaying an updated conceptual model that includes labelled coefficients for effects on crime rate. This figure was produced by using the package `semPlot` and directly inputs the regression results from *Table 1*. After presenting the direct, indirect, and total effects of each variable on crime rate, I recreated the initial conceptual model (*Figure 1*) and include the standardized regression coefficients for each variable, and present the effect results alongside this final conceptual model (*Figure 4*).

Overall, this analysis provides strong support for employing metropolitan status as a mediating variable on crime rate. This analysis eschews potentially racial and class-based prejudices that may be drawn from the relationship between crime rate and socioeconomic status. When incorporating metropolitan status as a mediating variable, the relationship of socioeconomic status and crime rate is complicated, and illuminates the complex nature of understanding crime in society.

Conceptual Path

In the below conceptual path (*Figure 1*), I illustrate the expected connections between the variables of interest. I expect that the % Hispanic and % in poverty will show direct, positive correlation with crime rate, along with metropolitan status. However, I expect metropolitan status to be a marked influence on crime rate within a county, and both % in poverty and % Hispanic will be mediated through metropolitan status. Therefore, five connecting lines are drawn to illustrate this relationship, and will be later tested in a series of regressions.

Figure 1: Conceptual Path

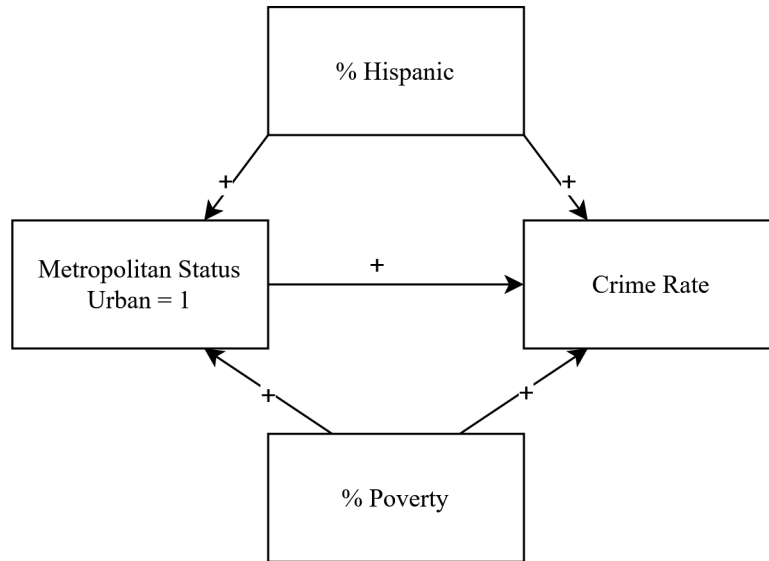
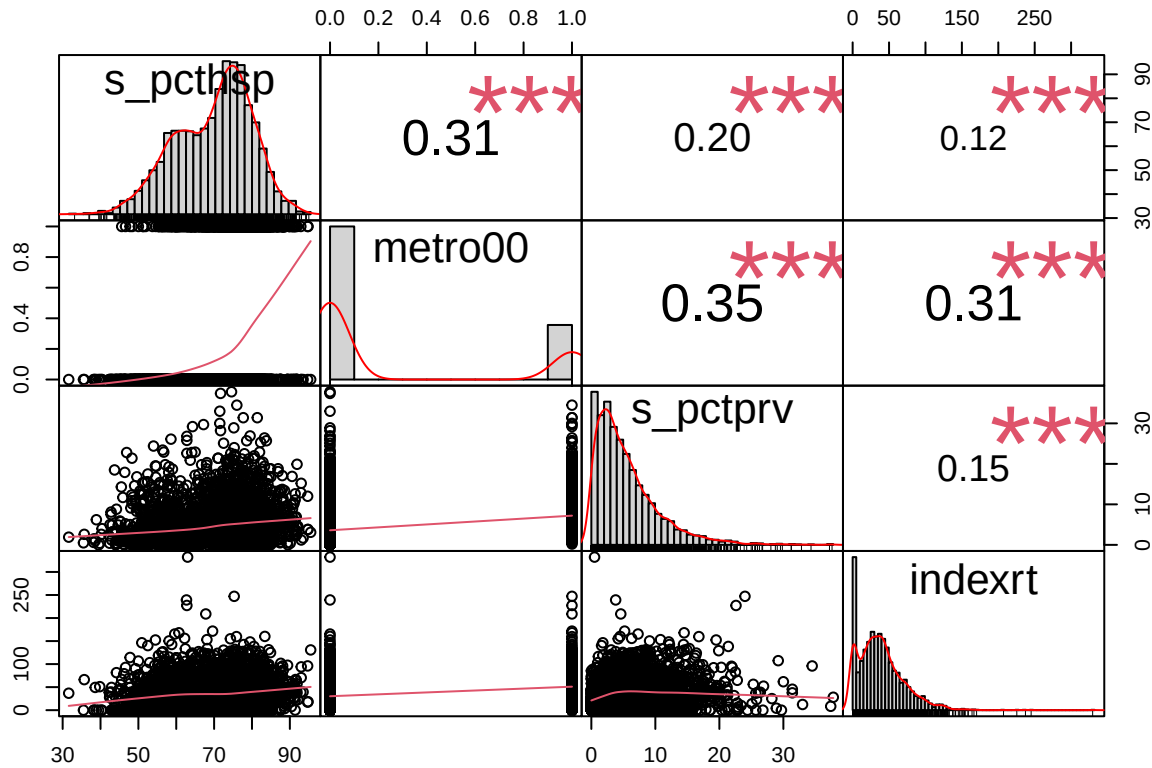


Figure 1: Conceptual Path

Correlation Matrix

The conceptual model drawn in *Figure 1* is supported by the below correlation matrix (*Figure 2*). A correlation matrix reveals that there are positive, statistically significant relationships between % Hispanic, % in poverty, and crime rate, as well as a positive, statistically significant relationship between metropolitan status and crime rate. This expected result justifies the inclusion of each variable in the conceptual model and further analysis. With metropolitan status having the largest correlation with not only crime rate, but % Hispanic, and % in poverty as well, this relationship shows support for considering metropolitan status as a mediating variable.

Figure 2: Correlation Matrix



Hypothesis Testing via Multiple Linear Regression

The below output is from a regression model that regresses the % Hispanic, % in poverty on the metropolitan status. This model is then followed by a second regression model that regresses the % Hispanic, % in poverty, and metropolitan status on the crime rate. Through this series of regressions, the mediating influence of metropolitan status can be revealed. Furthermore, the direct, indirect, and total effects of the variables of interest can be calculated. All-in-all, this regression model will test all five hypotheses of interest, and illuminate the complicated relationship between variables of interest on crime rate.

Regression Model

To test all five hypotheses, a series of multiple linear regression models is employed. The first model regresses % Hispanic (a) and % in poverty (d) on metropolitan status (c). This model is then followed by a second model that regresses % Hispanic (b), % in poverty (e), and metropolitan status (c) on crime rate ($indexrt$). Through this series of regressions, the mediating influence of metropolitan status can be revealed and the effects of % Hispanic and % in poverty can be disentangled.

Table 1: Regression Model Output

```

## Warning: lavaan->lav_data_full():
##   some observed variances are (at least) a factor 1000 times larger than
##   others; use varTable(fit) to investigate

## lavaan 0.6-21 ended normally after 1 iteration
##
##   Estimator                      ML
##   Optimization method          NLMINB
##   Number of model parameters      7
##
##                                     Used      Total
##   Number of observations          3067      3109
##
## Model Test User Model:
##
##   Test statistic                  0.000
##   Degrees of freedom              0
##
## Model Test Baseline Model:
##
##   Test statistic                  947.366
##   Degrees of freedom              5
##   P-value                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)      1.000
##   Tucker-Lewis Index (TLI)        1.000
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)      -16218.472
##   Loglikelihood unrestricted model (H1) -16218.472

```

```

##
## Akaike (AIC) 32450.944
## Bayesian (BIC) 32493.143
## Sample-size adjusted Bayesian (SABIC) 32470.901
##
## Root Mean Square Error of Approximation:
##
## RMSEA 0.000
## 90 Percent confidence interval - lower 0.000
## 90 Percent confidence interval - upper 0.000
## P-value H_0: RMSEA <= 0.050 NA
## P-value H_0: RMSEA >= 0.080 NA
##
## Standardized Root Mean Square Residual:
##
## SRMR 0.000
##
## Parameter Estimates:
##
## Standard errors Standard
## Information Expected
## Information saturated (h1) model Structured
##
## Regressions:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## metro00 ~
## s_pcthsp (a) 0.011 0.001 15.283 0.000 0.011 0.254
## s_pctprv (d) 0.027 0.001 18.172 0.000 0.027 0.302
## indexrt ~
## s_pcthsp (b) 0.077 0.054 1.437 0.151 0.077 0.026
## s_pctprv (e) 0.253 0.115 2.193 0.028 0.253 0.040
## metro00 (c) 20.095 1.326 15.150 0.000 20.095 0.288
##

```

```

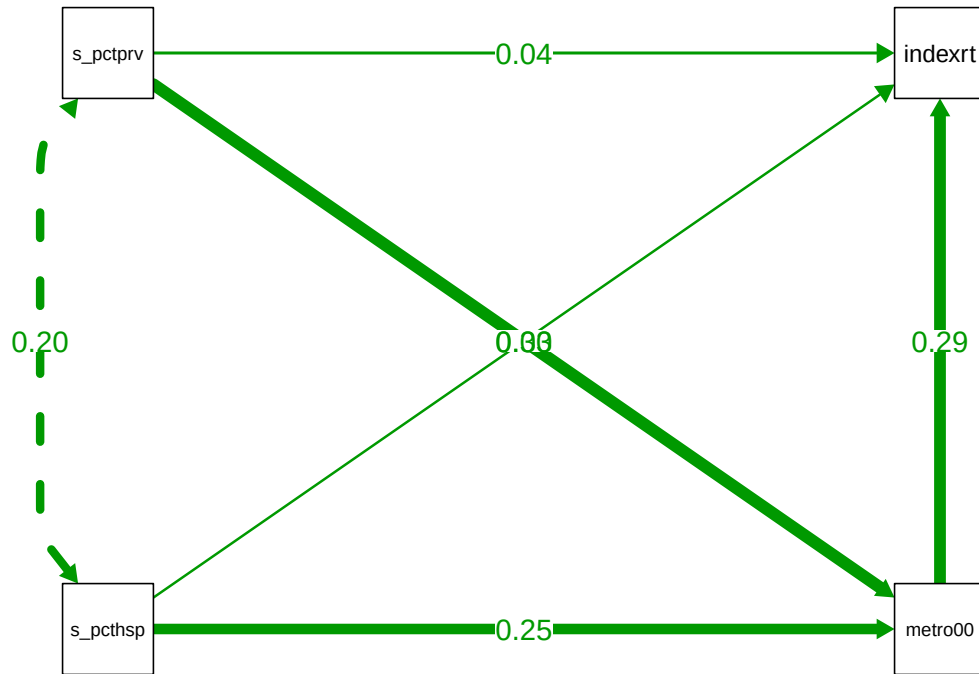
## Variances:
##
##          Estimate  Std.Err  z-value  P(>|z|)  Std.lv  Std.all
##    .metro00         0.158    0.004   39.160    0.000    0.158    0.814
##    .indexrt        851.420   21.742   39.160    0.000   851.420    0.902
##
## R-Square:
##
##          Estimate
##    metro00         0.186
##    indexrt         0.098
##
## Defined Parameters:
##
##          Estimate  Std.Err  z-value  P(>|z|)  Std.lv  Std.all
##    direct_hisp     0.077    0.054    1.437    0.151    0.077    0.026
##    direct_pov      0.253    0.115    2.193    0.028    0.253    0.040
##    indirect_hisp   0.217    0.020   10.759    0.000    0.217    0.073
##    indirect_pov    0.545    0.047   11.636    0.000    0.545    0.087
##    total_hisp      0.294    0.054    5.471    0.000    0.294    0.099
##    total_pov       0.798    0.114    7.021    0.000    0.798    0.127

```

Results Plot

The below figure (*Figure 3*) provides a visual representation of the regression results using the standardized coefficients of each variable, and their relationship to crime rate, as well as the mediating influence of metropolitan status.

Figure 3: Results Plot



Direct, Indirect, & Total Effects

The table below (*Table 2*) provides the direct, indirect, and total effects of the variables as a resulting from the regression model. After running the regression model, the direct effect of % Hispanic ($b^* = 0.026$) is no longer significant within this model. Therefore, we can conclude that the direct effect of % Hispanic on crime rate within a county is strongly mediated through the metropolitan status of a county. The % in poverty has a more significant indirect effect ($b^* = 0.087$, $p < 0.001$) via metropolitan status than the direct effect ($b^* = 0.040$, $p < 0.05$). While there is still a statistically significant positive association, but at a lower magnitude.

Table 2: Direct, Indirect, & Total Effects of Variables

##		Effect Estimate	Std_Estimate	P_value
## 1	Direct effect of % Hispanic	0.077	0.026	0.151
## 2	Direct effect of % Poverty	0.253	0.040	0.028
## 3	Indirect effect of % Hispanic (via Metro)	0.217	0.073	0.000
## 4	Indirect effect of % Poverty (via Metro)	0.545	0.087	0.000
## 5	Total effect of % Hispanic	0.294	0.099	0.000
## 6	Total effect of % Poverty	0.798	0.127	0.000

Conclusion

In this analysis I tested the following five hypotheses using a series of multiple linear regressions.

1. The % of a county's population identifying as Hispanic (**s_pcthsp**) predicts crime rate (**indexrt**) in a positive direction.
2. The metropolitan status of a county (**metro00 = 1**) predicts crime rate in a positive direction.
3. The % of a county's population in poverty (**s_pctprv**) predicts crime rate in a positive direction.
4. The % of a county's population identifying as Hispanic predicts metropolitan status in a positive direction.
5. The % of a county's population in poverty predicts metropolitan status in a positive direction.

The results of this analysis provide strong support for all five hypotheses. However, when investigating the direct and indirect effects of variables representing aspects of socioeconomic status, I found that the direct effect of the % Hispanic is not significant and the direct effect of poverty is less significant and stark when mediated through metropolitan status of a county. This finding provides support for further analyses to broaden a scope to incorporate other theoretically justified variables that may influence crime rate.

Below in *Figure 4* I recreate the initial conceptual model, but include the standardized coefficients of each variable, and present the direct, indirect (via metropolitan status), and total effects of % Hispanic and % in poverty on crime rate within a county.

Figure 4: Conceptual Model with Coefficients

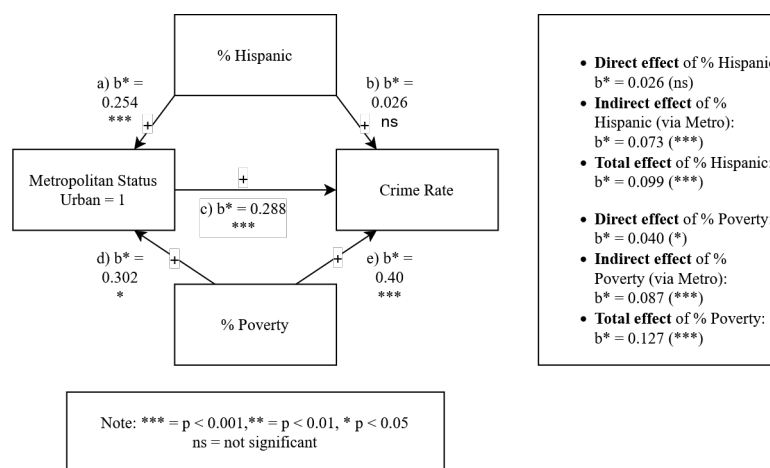


Figure 2: Conceptual Model with Coefficients